OAVS | TGT COMPUTER SCIENCE TEACHER EXAMINATION

Relational Algebra

MCQ Question Bank — Chapter 5

Select • Project • Union • Intersection • Difference • Cartesian Product • Simple Join • Query Solving

Total Questions: 160

Coverage: Strictly limited to Syllabus 5.1 (as per official topic list)

Sections: 13 (Introduction, seven core operations, comparison, query-solving, table-based, difficult/tricky, mixed practice)

Answer Key Distribution: A: 40 B: 40 C: 40 D: 40

Difficulty Mix: Easy / Moderate / Difficult–Tricky

Format: Each question includes 4 options, marked answer, and a short exam-oriented explanation

Prepared as an exam-preparation resource for OAVS TGT (Computer Science) aspirants.

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Section 1 — Introduction to Relational Algebra

Q1 Concept Easy

Relational Algebra is best described as a:

A. Procedural query language
B. Declarative query language
C. Object-oriented query language
D. Natural language interface

Ans: A

Explanation: Relational Algebra is procedural — a query specifies the exact sequence of operations needed to obtain the result.

Q2 Concept Easy

The result of every relational algebra operation is always a:

A. Single tuple
B. Single attribute
C. Domain value
D. Relation

Ans: D

Explanation: Relational algebra is closed under its operations — the output of any operation is itself a relation.

Q3 Concept Moderate

Which property of relational algebra allows the output of one operation to be used directly as the input of another operation?

A. Union compatibility
B. Closure property
C. Associative property
D. Reflexive property

Ans: B

Explanation: Because every operation returns a relation, operations can be nested or chained — this is the closure property.

Q4 Concept Easy

Among Select, Project, Union, Intersection, Difference, Cartesian Product and Simple Join, how many require exactly one input relation?

A. 1 (only Select)
B. 3 (Select, Project, Union)
C. 2 (Select and Project)
D. 0 (all need two relations)

Ans: C

Explanation: SELECT and PROJECT are unary operations; the remaining five operations in the syllabus each act on two relations.

Q5 Symbol Easy

Which symbol is used to denote the SELECT operation?

A. σ (sigma)
B. π (pi)
C. ∪ (union)
D. ⋈ (join)

Ans: A

Explanation: σ (sigma) is the standard symbol for the SELECT operation in relational algebra.

Q6 Symbol Easy

Which symbol is used to denote the PROJECT operation?

A. σ (sigma)
B. π (pi)
C. ∩ (intersection)
D. × (cross)

Ans: B

Explanation: π (pi) is the standard symbol used for the PROJECT operation.

Section 2 — SELECT Operation (σ)

Q7 Purpose Easy

What is the primary purpose of the SELECT operation?

A. To choose specific columns from a relation
B. To combine two relations
C. To choose tuples (rows) that satisfy a given condition
D. To remove duplicate tuples

Ans: C

Explanation: SELECT filters rows horizontally based on a condition; it does not touch columns.

Q8 Definition Easy

The SELECT operation in relational algebra is also known as a:

A. Duplication operation
B. Restriction operation
C. Aggregation operation
D. Extension operation

Ans: B

Explanation: SELECT restricts the relation to only those tuples meeting the condition, hence it is sometimes called restriction.

Q9 Syntax Moderate

The correct general syntax of the SELECT operation is:

A. π<condition>(Relation)
B. Relation σ <condition>
C. σ(Relation) <condition>
D. σ<condition>(Relation)

Ans: D

Explanation: The condition is written as a subscript to σ, followed by the relation name in parentheses.

Q10 Condition Moderate

In σ(Marks>75)(STUDENT), the expression 'Marks>75' is called the:

A. Join condition
B. Projection list
C. Selection condition (predicate)
D. Union condition

Ans: C

Explanation: The expression that a tuple must satisfy to be selected is called the selection condition or predicate.

Q11 Result Easy

The degree (number of attributes) of the relation produced by σ<condition>(R) compared to R is:

A. Same as R
B. Always less than R
C. Always greater than R
D. Always equal to 1

Ans: A

Explanation: SELECT only filters rows; it does not change the number of attributes, so the degree stays the same as R.

Q12 Result Moderate

The number of tuples in the result of σ<condition>(R) is:

A. Always equal to the number of tuples in R
B. Always greater than the number of tuples in R
C. Always exactly half of R
D. Less than or equal to the number of tuples in R

Ans: D

Explanation: SELECT can return zero, some, or all tuples of R, so the count is always ≤ the number of tuples in R.

Q13 Query-Solve Moderate

STUDENT(RollNo, Name, Marks): (1,Anil,78), (2,Sunil,45), (3,Kavita,92), (4,Rita,60). Result of σ(Marks>60)(STUDENT) contains how many tuples?

A. 2
B. 1
C. 3
D. 4

Ans: A

Explanation: Only Anil (78) and Kavita (92) satisfy Marks > 60, giving 2 tuples.

Q14 Query-Solve Moderate

Using the STUDENT relation above, σ(Marks<50)(STUDENT) returns tuples of:

A. Anil only
B. Sunil only
C. Rita only
D. Kavita and Rita

Ans: B

Explanation: Only Sunil has Marks = 45, which is less than 50.

Q15 Compound condition Moderate

Which logical operators can be used inside a SELECT condition to combine multiple predicates?

A. UNION and INTERSECT only
B. JOIN and PRODUCT only
C. AND (∧), OR (∨), NOT (¬)
D. GROUP BY and HAVING only

Ans: C

Explanation: Selection conditions may be combined using the logical connectives AND, OR and NOT.

Q16 Output-based Difficult

EMP(EmpId, Dept, Salary): (E1,IT,50000), (E2,HR,30000), (E3,IT,60000), (E4,Sales,40000). Result of σ(Dept='IT' AND Salary>55000)(EMP) is:

A. {(E1,IT,50000)}
B. {(E1,IT,50000),(E3,IT,60000)}
C. Empty relation
D. {(E3,IT,60000)}

Ans: D

Explanation: Only E3 satisfies both conditions simultaneously: Dept = 'IT' and Salary > 55000.

Q17 Interpretation Easy

σ(Class='XI')(STUDENT) selects:

A. All tuples where the Class attribute value equals 'XI'
B. All attributes named Class
C. Only the Class column
D. Tuples where Class is not 'XI'

Ans: A

Explanation: SELECT with an equality condition retrieves the rows whose specified attribute matches the given value.

Q18 Tricky Difficult

If a relation R has 10 tuples and σ<condition>(R) produces 10 tuples, what can be concluded?

A. The condition is invalid
B. R has duplicate tuples
C. The operation failed
D. Every tuple in R satisfies the condition

Ans: D

Explanation: If the output has the same number of tuples as the input, every tuple must have satisfied the selection condition.

Q19 Error-ID Difficult

Which of the following is an INCORRECT way to write a SELECT operation?

A. σ(Marks>60)(STUDENT)
B. σ(STUDENT)(Marks>60)
C. σ Marks>60 (STUDENT)
D. σ(Marks>60)(STUDENT) — written correctly with subscript

Ans: B

Explanation: The condition must appear as a subscript before the relation name in parentheses, not after it.

Q20 Assertion-Reason Difficult

Assertion (A): SELECT never changes the number of attributes of a relation. Reason (R): SELECT only filters tuples based on a condition. Choose the correct option.

A. A is true, R is false
B. A is false, R is true
C. Both A and R are true, and R correctly explains A
D. Both A and R are false

Ans: C

Explanation: Since SELECT filters rows only (based on R), the attribute count is unaffected — R correctly explains A.

Q21 Scenario Moderate

A school wants a list of all exam rows for students who scored below the passing mark. Which operation alone is sufficient?

A. SELECT
B. PROJECT
C. UNION
D. CARTESIAN PRODUCT

Ans: A

Explanation: Filtering rows by a condition (marks below passing) is precisely what the SELECT operation performs.

Q22 Query-Solve Difficult

Using STUDENT(RollNo,Name,Marks,Class) with 5 tuples where exactly 2 belong to Class 'XI', the result of σ(Class='XI')(STUDENT) has how many attributes and tuples?

A. 2 attributes, 4 tuples
B. 4 attributes, 5 tuples
C. 1 attribute, 2 tuples
D. 4 attributes, 2 tuples

Ans: D

Explanation: SELECT keeps all 4 attributes unchanged and returns only the 2 tuples matching Class = 'XI'.

Section 3 — PROJECT Operation (π)

Q23 Purpose Easy

The main purpose of the PROJECT operation is to:

A. Select specific rows from a relation
B. Select specific columns (attributes) from a relation
C. Combine two relations
D. Remove selected rows

Ans: B

Explanation: PROJECT works vertically — it extracts the chosen attributes from a relation.

Q24 Definition Easy

PROJECT is best described as an operation that produces a relation containing:

A. Only the specified rows of the original relation
B. Only the specified columns of the original relation
C. All rows and all columns unchanged
D. The intersection of two relations

Ans: B

Explanation: PROJECT builds a new relation using only the listed attributes from every tuple.

Q25 Syntax Moderate

The correct syntax for projecting the Name and Marks attributes from STUDENT is:

A. σ(Name, Marks)(STUDENT)
B. π(STUDENT)(Name, Marks)
C. π(Name, Marks)(STUDENT)
D. STUDENT π (Name, Marks)

Ans: C

Explanation: The attribute list is written as a subscript to π, followed by the relation in parentheses.

Q26 Duplicate handling Moderate

If projecting a single attribute produces repeated values across tuples, the PROJECT operation:

A. Keeps all duplicate values as separate tuples
B. Raises an error
C. Converts duplicates into NULL
D. Automatically removes the duplicate values from the result

Ans: D

Explanation: Since the result of PROJECT must be a valid relation (a set), duplicate tuples are eliminated automatically.

Q27 Result Easy

The degree of the relation resulting from π(A1,A2)(R) is:

A. 2, regardless of the degree of R
B. Same as the degree of R
C. Always 1
D. Equal to the number of tuples in R

Ans: A

Explanation: The degree of the projected relation equals the number of attributes listed in the projection, here 2.

Q28 Query-Solve Moderate

STUDENT(RollNo,Name,Class): (1,Anil,X),(2,Sunil,X),(3,Kavita,XI). Result of π(Class)(STUDENT) contains how many tuples?

A. 3
B. 1
C. 2
D. 4

Ans: C

Explanation: Projecting only Class gives 'X','X','XI'; the duplicate 'X' is removed, leaving 2 distinct tuples.

Q29 Query-Solve Moderate

Using the STUDENT relation above, π(Name,Class)(STUDENT) will contain how many tuples?

A. 3
B. 2
C. 1
D. 4

Ans: A

Explanation: Each (Name,Class) pair — (Anil,X),(Sunil,X),(Kavita,XI) — is distinct, so all 3 tuples remain.

Q30 Interpretation Easy

π(RollNo)(STUDENT) returns a relation with:

A. One row containing all RollNo values
B. All columns of STUDENT
C. Two columns, RollNo and Name
D. One column containing only distinct RollNo values

Ans: D

Explanation: PROJECT on a single attribute yields a single-column relation with distinct values only.

Q31 Output-based Difficult

EMP(EmpId,Dept,Salary) has 6 tuples spread across 3 distinct departments. π(Dept)(EMP) produces how many tuples?

A. 6
B. 3
C. 1
D. 0

Ans: B

Explanation: Even though there are 6 employee tuples, projecting only Dept leaves just the 3 distinct department values.

Q32 Combined Difficult

Which expression correctly retrieves the names of students scoring above 75 from STUDENT(RollNo,Name,Marks)?

A. π(Name)(σ(Marks>75)(STUDENT))
B. σ(Name)(π(Marks>75)(STUDENT))
C. π(Marks>75)(σ(Name)(STUDENT))
D. σ(Marks>75)(π(Name)(STUDENT))

Ans: A

Explanation: First filter rows with SELECT (Marks>75), then extract the Name column with PROJECT.

Q33 Error-ID Difficult

Which of these is NOT a valid reason to use PROJECT?

A. To reduce the number of attributes shown
B. To hide sensitive columns like Salary
C. To filter rows based on a numeric condition
D. To extract a list of distinct attribute values

Ans: C

Explanation: Filtering rows by a condition is the job of SELECT, not PROJECT.

Q34 Assertion-Reason Difficult

Assertion (A): π(A)(R) may contain fewer tuples than R. Reason (R): PROJECT eliminates duplicate tuples formed after removing unlisted attributes. Choose the correct option.

A. A is true, R is false
B. Both A and R are true, and R correctly explains A
C. A is false, R is true
D. Both A and R are false

Ans: B

Explanation: Dropping attributes can make previously distinct tuples identical; PROJECT then removes these duplicates, so A follows from R.

Q35 Scenario Moderate

A college wants only the list of subjects offered, without repetition, from a COURSE relation. Which operation is most appropriate?

A. SELECT
B. UNION
C. PROJECT
D. CARTESIAN PRODUCT

Ans: C

Explanation: Extracting one column with duplicates removed is exactly what PROJECT does.

Q36 Tricky Difficult

If π(A)(R) has the same number of tuples as R, what does this indicate?

A. R has no tuples
B. Attribute A is not part of R
C. R contains duplicate tuples
D. All values of attribute A in R are distinct

Ans: D

Explanation: No duplicates were removed, which is only possible if every tuple already had a unique value for A.

Q37 Query-Solve Moderate

STUDENT(RollNo,Name,Marks,Class) has 5 tuples. π(RollNo,Name)(STUDENT) has how many attributes and tuples?

A. 2 attributes, 5 tuples
B. 4 attributes, 5 tuples
C. 2 attributes, 2 tuples
D. 5 attributes, 2 tuples

Ans: A

Explanation: Since RollNo is a key, all 5 (RollNo,Name) pairs stay distinct, giving 2 attributes and 5 tuples.

Section 4 — UNION Operation (∪)

Q38 Purpose Easy

The UNION operation combines two relations to produce a relation containing:

A. Only tuples common to both relations
B. All tuples that appear in either relation (without duplicates)
C. Only tuples in the first relation but not the second
D. The pairwise combination of all tuples

Ans: B

Explanation: UNION returns the set of all distinct tuples present in R or S or both.

Q39 Compatibility Moderate

For R ∪ S to be valid, R and S must be:

A. Equal in number of tuples
B. Sorted in the same order
C. Union-compatible
D. Joined on a common key

Ans: C

Explanation: UNION requires union compatibility — same number of attributes with matching domains.

Q40 Compatibility Moderate

Two relations are union-compatible if they have:

A. The same primary key
B. At least one common attribute
C. The same relation name
D. The same number of attributes with corresponding attributes drawn from the same domain

Ans: D

Explanation: Union compatibility requires equal degree and matching attribute domains, not identical names or keys.

Q41 Duplicate handling Easy

If a tuple exists in both R and S, how many times does it appear in R ∪ S?

A. Once
B. Twice
C. Zero times
D. It depends on the order of the relations

Ans: A

Explanation: As a set operation, UNION automatically removes duplicate tuples, so it appears only once.

Q42 Query-Solve Moderate

R(RollNo,Name): (1,Anil),(2,Sunil). S(RollNo,Name): (2,Sunil),(3,Kavita). R ∪ S gives how many tuples?

A. 4
B. 3
C. 2
D. 1

Ans: B

Explanation: Combining {(1,Anil),(2,Sunil)} and {(2,Sunil),(3,Kavita)} and removing the duplicate (2,Sunil) leaves 3 tuples.

Q43 Result Easy

The degree of R ∪ S is:

A. Sum of the degrees of R and S
B. Equal to the number of tuples in R
C. Equal to the (same) degree of R and S
D. Always 1

Ans: C

Explanation: Since R and S must be union-compatible, R ∪ S has the same degree as R and S.

Q44 Interpretation Moderate

STUDENT_A holds students enrolled in Computer Science; STUDENT_B holds students enrolled in Mathematics. STUDENT_A ∪ STUDENT_B gives:

A. Students enrolled in both subjects only
B. Students enrolled in Computer Science or Mathematics (or both)
C. Students enrolled in Computer Science only
D. Students not enrolled in either subject

Ans: B

Explanation: UNION lists every student appearing in either relation, satisfying an 'or' condition.

Q45 Error-ID Difficult

Which pair of relations CANNOT be combined using UNION?

A. R(RollNo,Name) and S(Id,StudentName)
B. R(A,B) and S(C,D) with matching domains
C. Two copies of the same relation R
D. R(RollNo,Name) and S(EmpId,Dept,Salary)

Ans: D

Explanation: R has 2 attributes and S has 3, so they are not union-compatible and UNION cannot be applied.

Q46 Assertion-Reason Difficult

Assertion (A): R ∪ S is commutative, i.e. R ∪ S = S ∪ R. Reason (R): UNION treats relations as sets of tuples. Choose the correct option.

A. A is true, R is false
B. A is false, R is true
C. Both A and R are false
D. Both A and R are true, and R correctly explains A

Ans: D

Explanation: Because relations are sets (R justifies this), set union is commutative, making A true.

Q47 Scenario Moderate

A library wants a single combined list of members from its 'Book Section' and 'Journal Section' registers (same schema), with no member listed twice. Which operation is appropriate?

A. UNION
B. INTERSECTION
C. DIFFERENCE
D. CARTESIAN PRODUCT

Ans: A

Explanation: Merging two union-compatible relations while removing duplicates is exactly what UNION does.

Q48 Tricky Difficult

If R has 4 tuples and S has 5 tuples with 2 tuples common to both, how many tuples will R ∪ S contain?

A. 9
B. 2
C. 7
D. 4

Ans: C

Explanation: Total distinct tuples = 4 + 5 − 2 (common tuples counted once) = 7.

Q49 Query-Solve Moderate

R(Id,Name): (1,A),(2,B),(3,C). S(Id,Name): (4,D),(5,E). R ∪ S has how many tuples?

A. 5
B. 6
C. 3
D. 2

Ans: A

Explanation: R and S share no common tuples, so R ∪ S simply contains all 5 tuples from both relations.

Section 5 — INTERSECTION Operation (∩)

Q50 Purpose Easy

The INTERSECTION operation returns:

A. All tuples from both relations
B. Tuples present in R but not in S
C. The combination of all attribute pairs
D. Only the tuples common to both relations

Ans: D

Explanation: R ∩ S contains only those tuples that appear in both R and S.

Q51 Compatibility Moderate

INTERSECTION, like UNION, requires the two relations to be:

A. Joined on a foreign key
B. Union-compatible
C. Sorted identically
D. Of equal number of tuples

Ans: B

Explanation: R ∩ S is defined only when R and S are union-compatible (same degree, matching domains).

Q52 Result Easy

The degree of R ∩ S is:

A. Difference of the degrees of R and S
B. Always 1
C. Same as the degree of R and S
D. Sum of the degrees of R and S

Ans: C

Explanation: Since R and S are union-compatible, the result retains the same number of attributes.

Q53 Query-Solve Moderate

R(RollNo,Name): (1,Anil),(2,Sunil),(3,Kavita). S(RollNo,Name): (2,Sunil),(3,Kavita),(4,Rita). R ∩ S gives how many tuples?

A. 2
B. 3
C. 1
D. 4

Ans: A

Explanation: Only (2,Sunil) and (3,Kavita) are common to both R and S, giving 2 tuples.

Q54 Interpretation Moderate

STUDENT_SPORTS holds students in the Sports club; STUDENT_MUSIC holds students in the Music club. STUDENT_SPORTS ∩ STUDENT_MUSIC represents:

A. Students who are members of at least one club
B. Students who are members of only Sports club
C. Students who are not members of either club
D. Students who are members of both clubs

Ans: D

Explanation: INTERSECTION identifies tuples appearing in both relations — students in both clubs.

Q55 Error-ID Difficult

If R and S have no tuples in common, then R ∩ S is:

A. Equal to R
B. An empty relation (with the same schema as R and S)
C. Equal to S
D. Undefined

Ans: B

Explanation: With no common tuples, the intersection contains zero tuples but keeps the same schema.

Q56 Assertion-Reason Difficult

Assertion (A): R ∩ S can be expressed using DIFFERENCE as R − (R − S). Reason (R): Relational algebra operations can be equivalently rewritten using other operations. Choose the correct option.

A. A is true, R is false
B. Both A and R are true, and R correctly explains A
C. A is false, R is true
D. Both A and R are false

Ans: B

Explanation: R − (R − S) removes from R everything not in S, leaving exactly the tuples common to both — this is a valid equivalence, illustrating R.

Q57 Scenario Moderate

A school wants to find students who are enrolled in BOTH the Science and Commerce elective lists (same schema). Which operation should be used?

A. UNION
B. DIFFERENCE
C. INTERSECTION
D. CARTESIAN PRODUCT

Ans: C

Explanation: Finding tuples common to two relations is exactly what INTERSECTION performs.

Q58 Tricky Difficult

If R has 6 tuples, S has 6 tuples, and R ∩ S has 6 tuples, what can be concluded?

A. R and S share no tuples
B. S has more attributes than R
C. R is a subset of S only
D. R and S contain exactly the same set of tuples

Ans: D

Explanation: An intersection equal in size to both relations means every tuple of R is in S and vice versa — R = S.

Q59 Query-Solve Moderate

R(Id,City): (1,Delhi),(2,Mumbai),(3,Pune). S(Id,City): (4,Chennai),(5,Kolkata). R ∩ S contains:

A. No tuples (empty relation)
B. All 5 tuples
C. 3 tuples
D. 2 tuples

Ans: A

Explanation: R and S share no common tuples, so their intersection is empty.

Section 6 — DIFFERENCE Operation (−)

Q60 Purpose Easy

The DIFFERENCE operation R − S returns:

A. Tuples that are in S but not in R
B. Tuples that are in R but not in S
C. Tuples common to both R and S
D. All tuples from R and S combined

Ans: B

Explanation: R − S retains only the tuples of R that do not appear in S.

Q61 Order of operands Moderate

Is the DIFFERENCE operation commutative, i.e. does R − S = S − R always hold?

A. Yes, always
B. Only when R and S have equal tuples
C. No, DIFFERENCE is generally not commutative
D. Only when R and S are empty

Ans: C

Explanation: R − S and S − R generally give different results, since the order of operands changes which relation's exclusive tuples are kept.

Q62 Compatibility Moderate

DIFFERENCE also requires R and S to be:

A. Equal in number of tuples
B. Joined by a common attribute
C. Union-compatible
D. Sorted alphabetically

Ans: C

Explanation: Like UNION and INTERSECTION, DIFFERENCE requires union compatibility between R and S.

Q63 Query-Solve Moderate

R(RollNo,Name): (1,Anil),(2,Sunil),(3,Kavita). S(RollNo,Name): (2,Sunil). R − S gives:

A. {(1,Anil),(3,Kavita)}
B. {(2,Sunil)}
C. {(1,Anil),(2,Sunil),(3,Kavita)}
D. Empty relation

Ans: A

Explanation: R − S removes (2,Sunil), which is present in S, keeping the remaining two tuples of R.

Q64 Query-Solve Moderate

Using the same R and S above, S − R gives:

A. {(1,Anil),(3,Kavita)}
B. {(2,Sunil)}
C. {(1,Anil),(2,Sunil),(3,Kavita)}
D. Empty relation

Ans: D

Explanation: Every tuple of S, namely (2,Sunil), is also present in R, so S − R has no tuples left.

Q65 Interpretation Moderate

STUDENT_ALL holds all students; STUDENT_PASS holds students who passed. STUDENT_ALL − STUDENT_PASS represents:

A. Students who failed
B. Students who passed
C. All students, regardless of result
D. Students who both passed and failed

Ans: A

Explanation: Removing the passed students from the full list leaves exactly the students who did not pass.

Q66 Result Easy

The degree of R − S is:

A. Sum of the degrees of R and S
B. Always zero
C. Same as the degree of R and S
D. Difference of degrees of R and S

Ans: C

Explanation: Since R and S are union-compatible, R − S retains the same schema/degree.

Q67 Error-ID Difficult

If R and S are identical relations, R − S equals:

A. R itself
B. An empty relation
C. S itself
D. R ∪ S

Ans: B

Explanation: Removing all tuples of S (which equal all tuples of R) from R leaves nothing.

Q68 Assertion-Reason Difficult

Assertion (A): R − S is not commutative. Reason (R): DIFFERENCE keeps tuples exclusive to the first operand only. Choose the correct option.

A. A is true, R is false
B. Both A and R are true, and R correctly explains A
C. A is false, R is true
D. Both A and R are false

Ans: B

Explanation: Because DIFFERENCE depends on which relation is written first (R), swapping operands changes the result, confirming A.

Q69 Scenario Moderate

A college wants to find students registered for the Annual Exam but who did NOT appear for it (using two same-schema relations). Which operation is required?

A. UNION
B. INTERSECTION
C. CARTESIAN PRODUCT
D. DIFFERENCE

Ans: D

Explanation: Finding registered-but-not-appeared students means subtracting the 'appeared' relation from the 'registered' relation using DIFFERENCE.

Q70 Tricky Difficult

If R has 8 tuples and S has 3 tuples, all of which are also present in R, how many tuples will R − S contain?

A. 11
B. 3
C. 5
D. 8

Ans: C

Explanation: Removing the 3 common tuples from R's 8 tuples leaves 8 − 3 = 5 tuples.

Section 7 — CARTESIAN PRODUCT Operation (×)

Q71 Purpose Easy

The CARTESIAN PRODUCT operation R × S produces:

A. Every possible combination of tuples from R with tuples from S
B. Only matching tuples from R and S
C. Tuples present in R but not S
D. Tuples common to both relations

Ans: A

Explanation: R × S pairs each tuple of R with every tuple of S, producing all possible combinations.

Q72 Attribute count Easy

If R has 3 attributes and S has 4 attributes, the degree of R × S is:

A. 12
B. 1
C. 3
D. 7

Ans: D

Explanation: The degree of a Cartesian product equals the sum of the degrees of the two relations: 3 + 4 = 7.

Q73 Tuple count Moderate

If R has 4 tuples and S has 5 tuples, the number of tuples in R × S is:

A. 20
B. 9
C. 1
D. 5

Ans: A

Explanation: The number of tuples in a Cartesian product equals the product of the tuple counts: 4 × 5 = 20.

Q74 Combining tuples Moderate

In R × S, each resulting tuple is formed by:

A. Selecting the smaller tuple of the two
B. Concatenating one tuple of R with one tuple of S
C. Averaging attribute values of R and S
D. Merging only matching attributes

Ans: B

Explanation: CARTESIAN PRODUCT simply concatenates every tuple of R with every tuple of S, without any matching condition.

Q75 Query-Solve Moderate

R(RollNo): (1),(2). S(Subject): (Maths),(Science). R × S gives how many tuples?

A. 2
B. 1
C. 4
D. 6

Ans: C

Explanation: Each of R's 2 tuples is paired with each of S's 2 tuples: 2 × 2 = 4 tuples.

Q76 Query-Solve Difficult

Using R(RollNo) and S(Subject) from above, list the tuples of R × S.

A. (1,Maths),(2,Science)
B. (1,Maths),(1,Science),(2,Maths),(2,Science)
C. (1,Maths),(1,Science)
D. (1,2,Maths,Science)

Ans: B

Explanation: Every RollNo is combined with every Subject, producing all four ordered pairs.

Q77 Result Moderate

CARTESIAN PRODUCT is most commonly used in relational algebra as an intermediate step before applying:

A. A PROJECT operation only
B. A UNION operation only
C. A DIFFERENCE operation only
D. A SELECT condition to filter meaningfully related tuples

Ans: D

Explanation: R × S often produces many meaningless combinations; a SELECT condition is then applied to keep only related tuples — this underlies JOIN.

Q78 Compatibility Easy

Unlike UNION, INTERSECTION and DIFFERENCE, the CARTESIAN PRODUCT operation:

A. Requires the two relations to have the same degree
B. Requires the two relations to share a common attribute name
C. Does not require the two relations to be union-compatible
D. Requires the two relations to have equal tuples

Ans: C

Explanation: R × S can be applied to any two relations regardless of their schema; union compatibility is not required.

Q79 Error-ID Difficult

If R has 100 tuples and S has 200 tuples, computing R × S directly on large relations is generally considered:

A. Computationally expensive, as it may generate a very large number of tuples
B. Always the fastest way to combine data
C. Guaranteed to be more efficient than JOIN
D. Impossible to compute

Ans: A

Explanation: R × S here would generate 20,000 tuples, most likely unrelated, making it costly compared to a targeted JOIN.

Q80 Assertion-Reason Difficult

Assertion (A): SIMPLE JOIN can be expressed as a SELECT applied to a CARTESIAN PRODUCT. Reason (R): CARTESIAN PRODUCT generates all combinations, and SELECT can filter only the related ones. Choose the correct option.

A. Both A and R are true, and R correctly explains A
B. A is true, R is false
C. A is false, R is true
D. Both A and R are false

Ans: A

Explanation: σ<condition>(R × S) is exactly how a join is defined using the two more basic operations, so R explains A.

Q81 Scenario Moderate

A teacher wants to generate every possible pairing of Students with available Exam Slots before assigning actual slots. Which operation is suitable as the first step?

A. INTERSECTION
B. DIFFERENCE
C. PROJECT
D. CARTESIAN PRODUCT

Ans: D

Explanation: Generating all possible pairings between two relations is precisely the role of CARTESIAN PRODUCT.

Q82 Tricky Difficult

If R × S produces 42 tuples and R has 6 tuples, how many tuples does S have?

A. 36
B. 48
C. 7
D. 6

Ans: C

Explanation: Since |R × S| = |R| × |S|, S must have 42 ÷ 6 = 7 tuples.

Section 8 — SIMPLE JOIN Operation (⋈)

Q83 Definition Easy

SIMPLE JOIN is best described as a combination of:

A. UNION followed by PROJECT
B. CARTESIAN PRODUCT followed by a SELECT condition matching related attributes
C. INTERSECTION followed by DIFFERENCE
D. PROJECT followed by UNION

Ans: B

Explanation: A simple join first forms R × S and then applies a SELECT condition to keep only tuples where the join attributes match.

Q84 Purpose Easy

The main purpose of the SIMPLE JOIN operation is to:

A. Remove duplicate tuples from one relation
B. List all possible tuple combinations
C. Filter a single relation by a condition
D. Combine related tuples from two relations based on a common condition

Ans: D

Explanation: JOIN meaningfully combines rows from two relations that satisfy a specified matching condition.

Q85 Basic concept Moderate

In a simple join R ⋈(condition) S, the condition typically compares:

A. An attribute of R with an attribute of S
B. Two attributes of R only
C. Two attributes of S only
D. The degree of R with the degree of S

Ans: A

Explanation: The join condition relates attributes across the two relations, most commonly a foreign key to a primary key.

Q86 Syntax Moderate

R ⋈(R.DeptId = S.DeptId) S is equivalent to which combination of basic operations?

A. π(R.DeptId = S.DeptId)(R × S)
B. σ(R.DeptId = S.DeptId)(R × S)
C. σ(R.DeptId = S.DeptId)(R ∪ S)
D. π(DeptId)(R ∩ S)

Ans: B

Explanation: A simple join is defined as a SELECT with the join condition applied to the Cartesian product of R and S.

Q87 Matching tuples Easy

In SIMPLE JOIN, tuples appear in the result only if they:

A. Belong to both relations identically
B. Appear in either relation
C. Satisfy the specified join condition between the two relations
D. Have no attributes in common

Ans: C

Explanation: Only tuple pairs meeting the join condition are retained in the joined result.

Q88 Query-Solve Moderate

EMP(EmpId,Name,DeptId): (E1,Anil,D1),(E2,Sunil,D2). DEPT(DeptId,DeptName): (D1,IT),(D2,HR). EMP ⋈(EMP.DeptId=DEPT.DeptId) DEPT gives how many tuples?

A. 2
B. 4
C. 1
D. 0

Ans: A

Explanation: Each employee's DeptId matches exactly one department, so the join produces 2 matched tuples.

Q89 Query-Solve Moderate

Using EMP and DEPT above, the result of the join, when projected on Name and DeptName, would show:

A. (Anil, HR) and (Sunil, IT)
B. (Anil, IT) and (Anil, HR)
C. (Sunil, IT) and (Sunil, HR)
D. (Anil, IT) and (Sunil, HR)

Ans: D

Explanation: E1 (Anil) matches D1 (IT), and E2 (Sunil) matches D2 (HR), giving these two correctly matched pairs.

Q90 Result Moderate

Compared to R × S, the number of tuples in a simple join R ⋈ S is generally:

A. Always greater than R × S
B. Less than or equal to the number of tuples in R × S
C. Always equal to R × S
D. Always exactly half of R × S

Ans: B

Explanation: Since a join filters R × S using a matching condition, its result can never exceed the size of the full Cartesian product.

Q91 Error-ID Difficult

If no tuples in R and S satisfy the join condition, the result of R ⋈ S will be:

A. Equal to R × S
B. Equal to R
C. An empty relation
D. Undefined and an error occurs

Ans: C

Explanation: When no tuple pair satisfies the condition, SELECT filters out everything, leaving an empty relation.

Q92 Assertion-Reason Difficult

Assertion (A): SIMPLE JOIN reduces the number of unrelated tuple combinations compared to CARTESIAN PRODUCT. Reason (R): JOIN applies a condition that only related attribute values must satisfy. Choose the correct option.

A. A is true, R is false
B. A is false, R is true
C. Both A and R are false
D. Both A and R are true, and R correctly explains A

Ans: D

Explanation: Because R correctly states that the condition filters only matching pairs, A logically follows.

Q93 Scenario Moderate

A school wants a combined report showing each Student's Name along with their assigned Teacher's Name, using two relations linked by TeacherId. Which operation should be used?

A. SIMPLE JOIN
B. UNION
C. INTERSECTION
D. DIFFERENCE

Ans: A

Explanation: Combining related rows from two relations using a matching attribute (TeacherId) is exactly what SIMPLE JOIN performs.

Q94 Tricky Difficult

If R has 5 tuples, S has 6 tuples, and only 3 tuples of R have a matching DeptId in S, how many tuples will the simple join R ⋈ S contain (assuming each match is unique)?

A. 30
B. 11
C. 3
D. 5

Ans: C

Explanation: Only the 3 tuples of R with a genuine match in S survive the join condition, giving 3 result tuples.

Section 9 — Comparison of the Listed Operations

Q95 Comparison Easy

Which operation(s) act on a single relation?

A. UNION and INTERSECTION
B. SELECT and PROJECT
C. DIFFERENCE and CARTESIAN PRODUCT
D. SIMPLE JOIN and CARTESIAN PRODUCT

Ans: B

Explanation: SELECT and PROJECT are the only unary operations in the syllabus; the rest require two relations.

Q96 Comparison Moderate

Which of the following operations do NOT require union compatibility?

A. UNION and INTERSECTION
B. UNION and DIFFERENCE
C. INTERSECTION and DIFFERENCE
D. CARTESIAN PRODUCT and SIMPLE JOIN

Ans: D

Explanation: CARTESIAN PRODUCT and JOIN can combine relations of any schema, unlike UNION, INTERSECTION and DIFFERENCE.

Q97 Comparison Moderate

Which operation changes the number of attributes (degree) of the input relation(s)?

A. CARTESIAN PRODUCT (and SIMPLE JOIN)
B. SELECT
C. UNION
D. INTERSECTION

Ans: A

Explanation: CARTESIAN PRODUCT (and by extension JOIN) sums the degrees of the two relations, unlike SELECT, UNION or INTERSECTION.

Q98 Comparison Easy

Which operation is most closely associated with filtering rows using a logical condition?

A. PROJECT
B. UNION
C. SELECT
D. CARTESIAN PRODUCT

Ans: C

Explanation: SELECT is defined specifically to filter rows using a condition.

Q99 Comparison Easy

Which operation is most closely associated with filtering columns?

A. SELECT
B. PROJECT
C. INTERSECTION
D. DIFFERENCE

Ans: B

Explanation: PROJECT extracts a chosen subset of attributes (columns) from a relation.

Q100 Comparison Moderate

Which pair of operations both require the operand relations to be union-compatible AND both eliminate duplicate tuples automatically?

A. UNION and INTERSECTION
B. CARTESIAN PRODUCT and SIMPLE JOIN
C. SELECT and PROJECT
D. DIFFERENCE and CARTESIAN PRODUCT

Ans: A

Explanation: As set operations on union-compatible relations, both UNION and INTERSECTION inherently produce duplicate-free results.

Q101 Match-the-Following Moderate

Match: (i) σ (ii) π (iii) ∪ (iv) × with (a) columns (b) all combinations (c) rows (d) either-or membership. The correct matching is:

A. i-a, ii-c, iii-b, iv-d
B. i-b, ii-d, iii-a, iv-c
C. i-d, ii-b, iii-c, iv-a
D. i-c, ii-a, iii-d, iv-b

Ans: D

Explanation: σ filters rows (c), π extracts columns (a), ∪ combines with either-or membership (d), and × forms all combinations (b).

Q102 Comparison Difficult

Which operation, when applied, can NEVER increase the number of tuples beyond the larger of the two input relations?

A. CARTESIAN PRODUCT
B. UNION
C. SIMPLE JOIN
D. INTERSECTION

Ans: D

Explanation: INTERSECTION only keeps tuples common to both relations, so its size cannot exceed the smaller (and hence the larger) input's size.

Q103 Statement-based Moderate

Statement: 'This operation always requires the two relations to have an identical number of attributes and matching domains.' This statement is TRUE for:

A. CARTESIAN PRODUCT and SIMPLE JOIN only
B. UNION, INTERSECTION and DIFFERENCE
C. SELECT and PROJECT only
D. All seven operations

Ans: B

Explanation: Only UNION, INTERSECTION and DIFFERENCE demand union compatibility; the other operations do not.

Q104 Comparison Moderate

Which operation is used to derive a JOIN in relational algebra when no direct join operator is applied?

A. CARTESIAN PRODUCT combined with SELECT
B. PROJECT combined with UNION
C. INTERSECTION combined with DIFFERENCE
D. SELECT alone

Ans: A

Explanation: A join can always be simulated as σ<condition>(R × S), combining CARTESIAN PRODUCT with SELECT.

Q105 Tricky Difficult

For union-compatible relations R and S, which of the following relationships is always true: |R ∪ S| + |R ∩ S| = ?

A. |R| − |S|
B. |R| × |S|
C. |R| + |S|
D. |R| ÷ |S|

Ans: C

Explanation: This is the standard set-theory identity: the union size plus intersection size equals the sum of the individual sizes.

Q106 Comparison Easy

Which operation among the syllabus operations does NOT eliminate any duplicate tuples by definition (its schema simply reflects tuple pairing)?

A. UNION
B. CARTESIAN PRODUCT
C. INTERSECTION
D. DIFFERENCE

Ans: B

Explanation: CARTESIAN PRODUCT is a straightforward pairing of all tuples and does not perform any duplicate elimination step.

Section 10 — Query-Solving Questions

Q107 Query-Solve Moderate

BOOK(BookId,Title,Price): (B1,DBMS,300),(B2,OS,250),(B3,Networks,400). Result of σ(Price>280)(BOOK) contains:

A. B1 and B2
B. B2 and B3
C. B1 and B3
D. All three books

Ans: C

Explanation: Only DBMS (300) and Networks (400) have prices greater than 280.

Q108 Query-Solve Moderate

Using BOOK above, π(Title)(σ(Price<300)(BOOK)) gives:

A. DBMS
B. OS
C. Networks
D. DBMS and OS

Ans: B

Explanation: Only OS (250) has a price below 300; projecting its Title gives just 'OS'.

Q109 Query-Solve Moderate

TEACHER_A(TId,Name): (T1,Ravi),(T2,Meena). TEACHER_B(TId,Name): (T2,Meena),(T3,Suresh). Result of TEACHER_A ∪ TEACHER_B is:

A. {(T2,Meena)}
B. {(T1,Ravi),(T3,Suresh)}
C. {(T1,Ravi),(T2,Meena),(T2,Meena),(T3,Suresh)}
D. {(T1,Ravi),(T2,Meena),(T3,Suresh)}

Ans: D

Explanation: UNION merges both relations and removes the duplicate (T2,Meena), leaving 3 distinct tuples.

Q110 Query-Solve Moderate

Using TEACHER_A and TEACHER_B above, TEACHER_A ∩ TEACHER_B is:

A. {(T1,Ravi),(T3,Suresh)}
B. Empty relation
C. {(T2,Meena)}
D. {(T1,Ravi),(T2,Meena),(T3,Suresh)}

Ans: C

Explanation: Only (T2,Meena) is present in both relations, so the intersection contains just that tuple.

Q111 Query-Solve Moderate

Using TEACHER_A and TEACHER_B above, TEACHER_A − TEACHER_B is:

A. {(T1,Ravi)}
B. {(T3,Suresh)}
C. {(T2,Meena)}
D. Empty relation

Ans: A

Explanation: Removing the common tuple (T2,Meena) from TEACHER_A leaves only (T1,Ravi).

Q112 Query-Solve Difficult

STAFF(SId,Name): (S1,Rita),(S2,Mohan). ROOM(RoomNo): (101),(102),(103). The number of tuples in STAFF × ROOM is:

A. 5
B. 6
C. 2
D. 3

Ans: B

Explanation: |STAFF × ROOM| = |STAFF| × |ROOM| = 2 × 3 = 6.

Q113 Query-Solve Difficult

Using STAFF and ROOM above, the degree of STAFF × ROOM is:

A. 2
B. 1
C. 6
D. 3

Ans: D

Explanation: STAFF has 2 attributes and ROOM has 1, so the degree of the Cartesian product is 2 + 1 = 3.

Q114 Query-Solve Moderate

ORDER(OrderId,CustId): (O1,C1),(O2,C2). CUSTOMER(CustId,CustName): (C1,Anil),(C2,Sunil),(C3,Kavita). ORDER ⋈(ORDER.CustId=CUSTOMER.CustId) CUSTOMER gives how many tuples?

A. 2
B. 3
C. 6
D. 1

Ans: A

Explanation: Only C1 and C2 have matching orders, so exactly 2 tuples are produced; C3 (Kavita) has no matching order.

Q115 Query-Solve Difficult

Using ORDER and CUSTOMER above, π(OrderId,CustName)(ORDER ⋈(ORDER.CustId=CUSTOMER.CustId) CUSTOMER) gives:

A. (O1,Sunil) and (O2,Anil)
B. (O1,Anil),(O2,Sunil),(O3,Kavita)
C. (O1,Anil) and (O2,Sunil)
D. (O1,Anil) only

Ans: C

Explanation: O1 (C1) matches Anil and O2 (C2) matches Sunil; projecting OrderId and CustName gives these two correct pairs.

Q116 Query-Solve Moderate

STUDENT(RollNo,Name,Marks): (1,Anil,55),(2,Sunil,80),(3,Kavita,40),(4,Rita,90). π(Name)(σ(Marks>=80)(STUDENT)) results in:

A. {Sunil, Rita}
B. {Anil, Kavita}
C. {Sunil}
D. {Anil, Sunil, Rita}

Ans: A

Explanation: Sunil (80) and Rita (90) both satisfy Marks >= 80; projecting Name gives these two values.

Q117 Query-Solve Difficult

A(X,Y): (1,2),(3,4). B(X,Y): (1,2),(5,6). Which expression correctly gives tuples present ONLY in A, not in B?

A. B − A
B. A ∩ B
C. A ∪ B
D. A − B

Ans: D

Explanation: A − B removes any tuple of A that also appears in B, leaving tuples exclusive to A.

Q118 Query-Solve Difficult

Using A and B above, evaluate A − B.

A. {(1,2)}
B. {(3,4)}
C. {(1,2),(3,4)}
D. {(5,6)}

Ans: B

Explanation: The tuple (1,2) is common to both and is removed, leaving only (3,4) from A.

Q119 Query-Solve Difficult

EMPLOYEE(EId,DeptId,Salary): (E1,D1,40000),(E2,D2,55000),(E3,D1,60000). Result of π(EId)(σ(DeptId='D1' AND Salary>45000)(EMPLOYEE)) is:

A. {E1}
B. {E1,E3}
C. {E3}
D. {E1,E2,E3}

Ans: C

Explanation: Only E3 belongs to D1 and has Salary > 45000 (60000); the SELECT then PROJECT sequence yields just E3.

Q120 Query-Solve Difficult

R(A,B,C) has 5 tuples; S(D,E) has 3 tuples. If a join condition R.A=S.D is satisfied by exactly 2 tuple pairs, the degree and cardinality of R ⋈ S are:

A. Degree 2, Cardinality 5
B. Degree 5, Cardinality 2
C. Degree 5, Cardinality 15
D. Degree 15, Cardinality 2

Ans: B

Explanation: Degree = 3 + 2 = 5 (sum of attributes); cardinality = number of matching pairs = 2.

Q121 Query-Solve Moderate

CLASS_X(RollNo,Name): (1,Anil),(2,Sunil). CLASS_XI(RollNo,Name): (1,Anil),(3,Kavita). Which expression finds students common to both classes (possible only if RollNo is globally unique per student)?

A. CLASS_X − CLASS_XI
B. CLASS_X × CLASS_XI
C. CLASS_X ∪ CLASS_XI
D. CLASS_X ∩ CLASS_XI

Ans: D

Explanation: Finding tuples common to both relations is precisely the role of the INTERSECTION operation.

Q122 Query-Solve Difficult

Using CLASS_X and CLASS_XI above, evaluate CLASS_X ∩ CLASS_XI.

A. {(2,Sunil),(3,Kavita)}
B. Empty relation
C. {(1,Anil)}
D. {(1,Anil),(2,Sunil),(3,Kavita)}

Ans: C

Explanation: Only the tuple (1,Anil) is present in both relations.

Q123 Query-Solve Difficult

VEHICLE(VId,Type): (V1,Car),(V2,Bike),(V3,Car). π(Type)(VEHICLE) followed by counting the resulting tuples gives:

A. 2
B. 3
C. 1
D. 0

Ans: A

Explanation: The distinct Type values are 'Car' and 'Bike'; duplicates of 'Car' are removed, leaving 2 tuples.

Q124 Query-Solve Difficult

SUPPLIER(SId,City): (S1,Delhi),(S2,Mumbai). ITEM(SId,ItemName): (S1,Pen),(S3,Book). Result of SUPPLIER ⋈(SUPPLIER.SId=ITEM.SId) ITEM contains:

A. 2 tuples
B. Exactly 1 tuple, formed from S1's rows
C. 4 tuples
D. 0 tuples

Ans: B

Explanation: Only S1 exists in both relations, giving one matched tuple; S2 and S3 have no counterpart in the other relation.

Q125 Query-Solve Moderate

Which correctly expresses 'names of employees who work in the IT department', given EMPLOYEE(EId,Name,DeptId) and using only SELECT and PROJECT?

A. σ(Name)(π(DeptId='IT')(EMPLOYEE))
B. π(DeptId)(σ(Name='IT')(EMPLOYEE))
C. σ(DeptId='IT')(π(Name)(EMPLOYEE))
D. π(Name)(σ(DeptId='IT')(EMPLOYEE))

Ans: D

Explanation: First filter rows for DeptId = 'IT' using SELECT, then extract the Name column using PROJECT.

Q126 Query-Solve Difficult

Two relations R(A,B) and S(A,B) are union-compatible. If |R|=4, |S|=4 and R=S exactly, what is |R − S|?

A. 4
B. 8
C. 1
D. 0

Ans: D

Explanation: Since every tuple of R also exists in S, subtracting S from R removes everything, leaving zero tuples.

Section 11 — Table / Result-Based Questions

Q127 Table-based Moderate

Refer to the STUDENT table shown. How many tuples are returned by σ(Marks>=60)(STUDENT)?

RollNo	Name	Marks	Class
1	Anil	78	X
2	Sunil	45	X
3	Kavita	92	XI
4	Rita	60	XII
5	Mohan	38	XI

A. 3
B. 2
C. 4
D. 5

Ans: A

Explanation: Anil(78), Kavita(92) and Rita(60) satisfy Marks >= 60, giving 3 tuples.

Q128 Table-based Moderate

Refer to the STUDENT table shown. What is the result relation of π(Class)(STUDENT)?

RollNo	Name	Marks	Class
1	Anil	78	X
2	Sunil	45	X
3	Kavita	92	XI
4	Rita	60	XII
5	Mohan	38	XI

A. {X, XI, XII} — 3 distinct tuples
B. {X, X, XI, XII, XI} — 5 tuples
C. {X, XI} — 2 tuples
D. {XII} — 1 tuple

Ans: A

Explanation: The distinct Class values across all 5 tuples are X, XI and XII, giving 3 tuples after duplicate removal.

Q129 Table-based Difficult

Refer to the STUDENT table shown. How many tuples satisfy σ(Class='XI' AND Marks>40)(STUDENT)?

RollNo	Name	Marks	Class
1	Anil	78	X
2	Sunil	45	X
3	Kavita	92	XI
4	Rita	60	XII
5	Mohan	38	XI

A. 2
B. 0
C. 1
D. 3

Ans: C

Explanation: Class 'XI' has Kavita(92) and Mohan(38); only Kavita also satisfies Marks > 40.

Q130 Table-based Moderate

Refer to the STUDENT table shown. The result of π(Name,Marks)(σ(Class='X')(STUDENT)) is:

RollNo	Name	Marks	Class
1	Anil	78	X
2	Sunil	45	X
3	Kavita	92	XI
4	Rita	60	XII
5	Mohan	38	XI

A. {(Kavita,92),(Rita,60)}
B. {(Anil,78)}
C. {(Anil,78),(Sunil,45),(Rita,60)}
D. {(Anil,78),(Sunil,45)}

Ans: D

Explanation: Only Anil and Sunil belong to Class 'X'; projecting Name and Marks gives these two tuples.

Q131 Table-based Difficult

Refer to the DEPT and EMP tables shown. How many tuples result from EMP ⋈(EMP.DeptId=DEPT.DeptId) DEPT?

DeptId	DeptName
D1	IT
D2	HR

EmpId	Name	DeptId
E1	Anil	D1
E2	Sunil	D2
E3	Kavita	D1

A. 2
B. 3
C. 6
D. 1

Ans: B

Explanation: Every employee's DeptId (D1 or D2) matches a department, so all 3 employee tuples find a match.

Q132 Table-based Moderate

Refer to the DEPT and EMP tables shown. How many tuples are in EMP × DEPT?

DeptId	DeptName
D1	IT
D2	HR

EmpId	Name	DeptId
E1	Anil	D1
E2	Sunil	D2
E3	Kavita	D1

A. 3
B. 5
C. 6
D. 2

Ans: C

Explanation: |EMP × DEPT| = |EMP| × |DEPT| = 3 × 2 = 6.

Q133 Table-based Difficult

Refer to the DEPT and EMP tables shown. π(Name)(σ(DeptId='D1')(EMP)) gives:

DeptId	DeptName
D1	IT
D2	HR

EmpId	Name	DeptId
E1	Anil	D1
E2	Sunil	D2
E3	Kavita	D1

A. {Anil, Kavita}
B. {Sunil}
C. {Anil, Sunil}
D. {Anil, Sunil, Kavita}

Ans: A

Explanation: Anil and Kavita both have DeptId 'D1'; projecting Name gives these two values.

Q134 Table-based Moderate

Refer to the two relations R (top) and S (bottom) shown. How many tuples does R ∪ S contain?

Id	Name
1	Aman
2	Priya
3	Rohit

Id	Name
2	Priya
3	Rohit
4	Zoya

A. 6
B. 3
C. 2
D. 4

Ans: D

Explanation: Combining R and S and removing the two common tuples (2,Priya) and (3,Rohit) leaves 4 distinct tuples.

Q135 Table-based Moderate

Refer to R and S shown above. How many tuples does R ∩ S contain?

Id	Name
1	Aman
2	Priya
3	Rohit

Id	Name
2	Priya
3	Rohit
4	Zoya

A. 1
B. 2
C. 3
D. 0

Ans: B

Explanation: The tuples (2,Priya) and (3,Rohit) are common to both R and S.

Q136 Table-based Moderate

Refer to R and S shown above. What is R − S?

Id	Name
1	Aman
2	Priya
3	Rohit

Id	Name
2	Priya
3	Rohit
4	Zoya

A. {(4,Zoya)}
B. {(1,Aman),(4,Zoya)}
C. {(1,Aman)}
D. Empty relation

Ans: C

Explanation: Removing tuples common to S from R leaves only (1,Aman), which is exclusive to R.

Q137 Table-based Difficult

Refer to R and S shown above. What is S − R?

Id	Name
1	Aman
2	Priya
3	Rohit

Id	Name
2	Priya
3	Rohit
4	Zoya

A. {(4,Zoya)}
B. {(1,Aman)}
C. {(1,Aman),(4,Zoya)}
D. Empty relation

Ans: A

Explanation: Removing tuples common to R from S leaves only (4,Zoya), which is exclusive to S.

Section 12 — Difficult & Tricky Questions

Q138 Tricky Difficult

Which single relational algebra expression is equivalent to 'tuples in R that are NOT related to any tuple in S', assuming R and S are union-compatible?

A. R ∩ S
B. R − S
C. R × S
D. R ⋈ S

Ans: B

Explanation: DIFFERENCE directly isolates tuples of R having no counterpart in S.

Q139 Tricky Difficult

If R ∩ S is empty, but R and S are union-compatible, then R ∪ S will always have how many tuples?

A. |R| − |S|
B. |R| × |S|
C. |R| + |S|
D. 0

Ans: C

Explanation: With no overlap, the union simply combines all tuples from both relations, giving |R| + |S|.

Q140 Tricky Difficult

Given R(A,B,C) with 10 tuples, which of the following expressions is guaranteed to never exceed 10 tuples in its result?

A. R × R
B. σ<condition>(R)
C. R × S (for any non-empty S)
D. π(A)(R) ∪ π(B)(R)

Ans: B

Explanation: SELECT only filters R's own tuples, so its result can never exceed R's original 10 tuples; the other expressions can produce more.

Q141 Assertion-Reason Difficult

Assertion (A): π(A,B)(R) can have fewer attributes than R but the same or fewer number of tuples. Reason (R): Projecting may cause previously distinct tuples to become identical, triggering duplicate elimination. Choose the correct option.

A. A is true, R is false
B. A is false, R is true
C. Both A and R are false
D. Both A and R are true, and R correctly explains A

Ans: D

Explanation: R correctly explains why the tuple count in A can drop after projection — duplicates created by dropping attributes are removed.

Q142 Error-ID Difficult

Identify the incorrect statement about SIMPLE JOIN.

A. Simple join can be performed on relations that are not union-compatible only if their degrees are equal
B. Simple join does not require union compatibility
C. Simple join can be expressed as σ(condition)(R × S)
D. Simple join filters unrelated tuple combinations produced by Cartesian product

Ans: A

Explanation: SIMPLE JOIN never requires union compatibility or equal degree — it only needs a valid join condition relating attributes of R and S.

Q143 Tricky Difficult

R(A,B) and S(A,B) are union-compatible with |R|=7 and |S|=5. If |R ∪ S|=9, what is |R ∩ S|?

A. 2
B. 5
C. 3
D. 12

Ans: C

Explanation: Using |R ∪ S| = |R| + |S| − |R ∩ S|: 9 = 7 + 5 − |R ∩ S|, so |R ∩ S| = 3.

Q144 Tricky Difficult

If R has 4 attributes and S has 3 attributes, and R ⋈ S is computed using exactly one equality join condition, the degree of the result is:

A. 7
B. 6
C. 3
D. 4

Ans: A

Explanation: A simple join (unlike a natural join) does not remove the duplicate join column, so the degree remains 4 + 3 = 7.

Q145 Scenario Difficult

A result relation must list students who are enrolled in the Science elective but NOT enrolled in the Commerce elective (both relations have identical schema). Which single operation directly gives this result?

A. INTERSECTION
B. UNION
C. CARTESIAN PRODUCT
D. DIFFERENCE (Science − Commerce)

Ans: D

Explanation: DIFFERENCE precisely isolates tuples present in the Science relation but absent from the Commerce relation.

Q146 Tricky Difficult

For relations R and S, which of the following expressions is generally NOT equal to R ∩ S (even though it looks similar)?

A. R − (R − S)
B. R × S
C. S − (S − R)
D. Both B and C are valid equivalences of R ∩ S

Ans: B

Explanation: CARTESIAN PRODUCT combines all tuple pairs and changes the degree entirely; it is not equivalent to INTERSECTION.

Q147 Assertion-Reason Difficult

Assertion (A): The order of applying SELECT and PROJECT together can affect efficiency but not the final result, when filtering rows then extracting unrelated columns. Reason (R): SELECT and PROJECT operate independently on rows and columns respectively when the selection attribute is retained in the projection list. Choose the correct option.

A. A is true, R is false
B. A is false, R is true
C. Both A and R are true, and R correctly explains A
D. Both A and R are false

Ans: C

Explanation: As long as the selection attribute remains available, filtering before or after projecting gives the same final tuples, which R explains.

Q148 Tricky Difficult

R(A,B,C,D) has 20 tuples. S = π(A,B)(R). If S has 20 tuples, what does this imply about attributes A and B in R?

A. A and B are both NULL
B. R has duplicate tuples
C. C and D are redundant attributes
D. The combination of (A,B) is unique across all tuples of R

Ans: D

Explanation: No tuples were merged during projection, meaning every (A,B) pair in R was already distinct.

Section 13 — Mixed OAVS TGT Practice

Q149 Mixed Easy

Which relational algebra operation retrieves specific rows satisfying a Boolean condition?

A. SELECT
B. PROJECT
C. UNION
D. CARTESIAN PRODUCT

Ans: A

Explanation: SELECT (σ) is defined to filter rows according to a given condition.

Q150 Mixed Easy

Which operation is used to eliminate unwanted columns from a relation?

A. SELECT
B. PROJECT
C. INTERSECTION
D. DIFFERENCE

Ans: B

Explanation: PROJECT (π) retains only the specified attributes, effectively removing the rest.

Q151 Mixed Moderate

Two relations R(RollNo,Name) and S(RollNo,Name,Marks) are given. Which operation CANNOT be directly applied between them without modification?

A. CARTESIAN PRODUCT
B. SIMPLE JOIN (on RollNo)
C. UNION (they are not union-compatible)
D. None — all operations work directly

Ans: C

Explanation: R has 2 attributes and S has 3, so they fail union compatibility and UNION cannot be applied directly.

Q152 Mixed Moderate

Which operation combines tuples from two relations without requiring any matching condition, potentially producing meaningless combinations?

A. CARTESIAN PRODUCT
B. SIMPLE JOIN
C. INTERSECTION
D. DIFFERENCE

Ans: A

Explanation: CARTESIAN PRODUCT pairs every tuple of one relation with every tuple of another, regardless of relevance.

Q153 Mixed Difficult

R(A,B) has 3 tuples, S(A,B) has 3 tuples, and R = S. What is the result of R ∪ S, R ∩ S and R − S respectively?

A. Empty, Empty, Empty
B. R (3 tuples), Empty, R (3 tuples)
C. R (6 tuples), R (3 tuples), Empty
D. R (3 tuples), R (3 tuples), Empty relation

Ans: D

Explanation: Since R = S: their union equals R (duplicates removed), their intersection equals R, and their difference is empty.

Q154 Mixed Moderate

Which of the following best explains why CARTESIAN PRODUCT is considered a 'foundation' operation for JOIN?

A. Because JOIN always produces fewer attributes than CARTESIAN PRODUCT
B. Because CARTESIAN PRODUCT requires union compatibility like JOIN
C. Because JOIN can be defined as SELECT applied over the result of CARTESIAN PRODUCT
D. Because JOIN and CARTESIAN PRODUCT always produce identical results

Ans: C

Explanation: SIMPLE JOIN is formally defined as σ<condition>(R × S), making CARTESIAN PRODUCT its foundation.

Q155 Mixed Easy

The number of attributes in a relation is also called its:

A. Cardinality
B. Degree
C. Domain
D. Tuple count

Ans: B

Explanation: Degree refers to the number of attributes; cardinality refers to the number of tuples.

Q156 Mixed Moderate

Given STUDENT(RollNo,Name,Marks) and RESULT(RollNo,Grade), which operation would you use to list each student's Name alongside their Grade?

A. SIMPLE JOIN on RollNo
B. UNION
C. INTERSECTION
D. DIFFERENCE

Ans: A

Explanation: Combining related tuples from two relations using a common attribute (RollNo) is the role of SIMPLE JOIN.

Q157 Mixed Difficult

R(A,B,C) has 8 tuples. π(A,B)(R) has 6 tuples. What does this tell us?

A. R had exactly 6 unique values of C
B. Attribute C is a key of R
C. The projection operation failed
D. At least two tuples had identical (A,B) values, which merged after projection

Ans: D

Explanation: Since the tuple count dropped from 8 to 6, some (A,B) combinations must have repeated and merged during projection.

Q158 Mixed Difficult

Which of the following correctly orders the operations to answer: 'List names of Class XI students scoring above 80', given STUDENT(RollNo,Name,Marks,Class)?

A. Apply PROJECT first (Name), then apply SELECT (Class='XI' AND Marks>80)
B. Apply UNION first, then SELECT
C. Apply SELECT first (Class='XI' AND Marks>80), then apply PROJECT (Name)
D. Apply CARTESIAN PRODUCT first, then PROJECT

Ans: C

Explanation: PROJECT must be applied last here, since projecting Name first would remove the Class and Marks attributes needed for the condition.

Q159 Mixed Moderate

Which of these operations is idempotent, meaning applying it twice with the same condition/attributes gives the same result as applying it once?

A. Only CARTESIAN PRODUCT
B. Both SELECT and PROJECT
C. Only UNION
D. Only DIFFERENCE

Ans: B

Explanation: Re-applying the same SELECT condition or the same PROJECT attribute list again changes nothing further — both are idempotent.

Q160 Mixed Difficult

An OAVS TGT exam question states: 'R has 12 tuples; S has 12 tuples; R ∪ S has 12 tuples.' What must be true?

A. R and S are disjoint
B. S has more attributes than R
C. R ∩ S is empty
D. R and S contain exactly the same set of tuples

Ans: D

Explanation: For the union to still have only 12 tuples (not more), every tuple of S must already exist in R, meaning R and S are identical sets.